

Phases 6, 7, 8 – Easy → Hard (1 CR Product-Company Guide)

MERN MASTERY · 1 CR PRODUCT PREP · HARSH VARDHAN GUPTA

 **Study window:** 7 Dec 2026 – 14 Feb 2027 (Days 113–182)

 **Your slots (IST):** Mon–Fri **7:30–8:30pm** & **9:00–10:00pm** · Sat–Sun **9:00am–12:00pm**, **3:00–5:30pm**, **6:00–8:30pm**

 **Calendar reminders:** harsh.gupta@getreelax.com – import `calendar/mern-mastery-study.ics`

Purpose: Phase 6–8 is where a **~1 CR product-company** offer is won or lost. Every sub-phase builds on the previous. **Never skip a sub-phase.**

Companion: `20-1cr-product-company-complete.pdf` — company loops, question types, product HLD, frontend MC, CS pack, optional months 7–9.

Overview

PHASE 6 (Days 113–140) – DSA + LLD (product-company)

- 6A  Easy DSA patterns
- 6B  Easy LLD / OOP
- 6C  Medium DSA trees & graphs
- 6D  Medium LLD (LRU, Rate Limiter)
- 6E  Medium-Hard DSA (DP, intervals, heap)
- 6F  Hard + company-tagged DSA
- 6G  Backend MC + frontend MC

PHASE 7 (Days 141–168) – Product system design

- 7A  Easy SD building blocks
- 7B  Easy HLD (single-service designs)
- 7C  Medium product HLD (feeds, notifications)
- 7D  Medium-Hard product HLD (chat, payments, Uber)
- 7E  Hard distributed (product-framed)
- 7F  Product HLD deep dives (YouTube, Drive, Typeahead)
- 7G  CS fundamentals + FE MC + company loops

PHASE 8 (Days 169–182) – Interview execution

- 8A  Easy review + resume
- 8B  Medium partial mocks
- 8C  Medium-Hard combined mocks
- 8D  Hard full loop (company-shaped)
- 8E  Hard apply + negotiate

PHASE 9 (optional Days 183–273) – L5 / 1 CR+ polish

See 20-1cr-product-company-complete.pdf §11

Legend:  Easy |  Medium |  Medium-Hard |  Hard (1 CR product bar)

PHASE 6 – DSA + LLD (Days 113-140)

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Goal: Medium LC in 30 min → company-tagged + follow-ups → 90 min backend MC + one frontend widget

1 CR bar: Google L4 / Meta E4 / Amazon SDE2 = 2 mediums (one with follow-up) + product communication. L5 adds a hard or a design-LC.

6A 🟢 EASY – DSA pattern refresh (Days 113-116)

Difficulty: Easy → Easy-Medium LeetCode

Theory: [08-dsa-patterns-complete.pdf](#) §1-4, §10 (stack)

Review: [08b-dsa-datastructures-fundamentals.pdf](#) §1-5

Day	Topic	LeetCode	Time limit
113	Two pointers review	LC 167 Two Sum II, LC 125 Valid Palindrome	25 min each
114	Hash map review	LC 1 Two Sum, LC 242 Valid Anagram	20 min each
115	Stack review	LC 20 Valid Parentheses, LC 155 Min Stack	25 min each
116	Sliding window intro	LC 121 Best Time Buy/Sell, LC 283 Move Zeroes	25 min each

What success looks like: Solve easy-medium in 25 min without hints. Explain pattern name aloud.

Do NOT proceed to 6B until: 4/4 days completed, all solutions without looking at notes.

6B 🟢 EASY – LLD / OOP fundamentals (Days 117-119)

Difficulty: Simple class design, no concurrency yet

Theory: [09-lld-complete.pdf](#) §SOLID, §Design Patterns intro

Interview Q&A: Tier 3 LLD Q1-5 ([17-interview-qa](#) §7)

Day	Build	Focus	Time
117	Snake & Ladder (basic)	Classes: Board, Player, Dice; no threading	2 hrs
118	Vending Machine	State pattern: Idle, HasMoney, Dispensing	2 hrs
119	SOLID quiz + draw Parking Lot class diagram only (no code yet)	Entities: Vehicle, Spot, Floor, Ticket	2 hrs

What success looks like: Draw class diagram in 10 min. Explain Single Responsibility for each class.

Gate: Name 5 SOLID principles with one example each.

6C 🟡 MEDIUM – DSA trees & graphs (Days 120-123)

Difficulty: Medium LeetCode

Theory: [08-dsa-patterns-complete.pdf](#) §5–9, [08b](#) §6–8

Day	Topic	LeetCode	Time limit
120	Tree traversals	LC 94 Inorder, LC 102 Level Order	30 min each
121	Tree properties	LC 104 Max Depth, LC 226 Invert Tree	25 min each
122	Graph BFS	LC 200 Number of Islands	35 min
123	Graph DFS + topo	LC 207 Course Schedule	35 min

What success looks like: BFS/DFS template from memory. Medium in 35 min.

6D 🟡 MEDIUM – LLD standard problems (Days 124-127)

Difficulty: 45-min LLD rounds (Flipkart, Amazon SDE2)

Theory: [09-lld-complete.pdf](#) + [15-machine-coding-complete.pdf](#)

Day	Build	Pattern	Time
124	LRU Cache	HashMap + Doubly Linked List	45 min timed
125	Rate Limiter	Sliding window / token bucket	45 min timed
126	Parking Lot (full code)	Factory + Strategy (vehicle types)	60 min
127	Review + redo weakest LLD blind	–	2 hrs

What success looks like: LRU Cache coded in 35 min with working get/put $O(1)$.

Gate: 2 LLD problems coded blind before Phase 6E.

6E 🟡 MEDIUM-HARD – Advanced DSA patterns (Days 128-132)

Difficulty: Medium-Hard LeetCode

Theory: [08-dsa-patterns-complete.pdf](#) §8 (DP), [14-advanced-dsa-hard.pdf](#) §5 intro only

Day	Topic	LeetCode	Time limit
128	1D DP	LC 70 Climbing Stairs, LC 198 House Robber	30 min each
129	DP patterns	LC 322 Coin Change, LC 139 Word Break	40 min each
130	Binary search on answer	LC 33 Search Rotated, LC 875 Koko Bananas	35 min each
131	Heap + intervals	LC 347 Top K, LC 56 Merge Intervals, LC 253 Meeting Rooms II	35 min each
132	Monotonic stack + backtracking	LC 739 Daily Temps, LC 78 Subsets	40 min

Also read: [08-dsa-patterns](#) §16–18 (product-company DSA, intervals, backtracking). After each problem, speak one follow-up ([20-1cr](#) §4.3).

What success looks like: Recognize DP / interval / backtracking in 5 min. 150+ total LC.

6F 🟡 HARD – Advanced DSA (Days 133-135)

Difficulty: Hard + company-tagged mediums (1 CR coding bar)

Theory: [14-advanced-dsa-hard-complete.pdf](#) (tries, UF, topo, DP — skip segment trees unless extra time) • [20-1cr](#) §4.2

Day	Topic	LeetCode	Time limit
133	Hard arrays + Meta speed	LC 42 Trapping Rain, LC 76 Min Window, LC 560 Subarray Sum K	40 min each
134	Hard graphs/DP + Amazon	LC 124 Max Path Sum, LC 200 Islands (blind redo), LC 994 Rotting Oranges	40 min
135	Design-LC + Uber/Google	LC 146 LRU (blind), LC 23 Merge K or LC 295 Median Stream	45 min

What success looks like: Attempt hard even if not optimal in 45 min — then speak a stream/memory follow-up. Pick your **first target company** tagged set from [20-1cr](#) §4.2 and start it.

Gate: 200+ total LC before Phase 7.

6G 🟡 HARD – Machine coding 90 min (Days 136-140)

Difficulty: Full 90-min rounds (Amazon, Flipkart, Uber, Atlassian)

Theory: [15-machine-coding-complete.pdf](#) + §13 frontend MC

Day	Build	Requirements	Time
136	Splitwise	Equal/exact split, settle up, print balances	90 min strict
137	BookMyShow	Seat map, booking, lock seat 5 min	90 min strict
138	Elevator	Multiple elevators, direction, floor requests	90 min strict
139	Frontend MC: Autocomplete	Debounce, keyboard, abort fetch, empty/loading (15 §13)	90 min strict
140	Phase 6 gate exam	1 medium LC (35 min) + 1 LLD (45 min) + review	3 hrs

Phase 6 final gate (must pass ALL):

- ☐ 200+ LeetCode (120+ medium)
- ☐ Medium LC in 30 min average + one spoken follow-up
- ☐ Intervals + backtracking + Union-Find: 2 problems each blind
- ☐ 6 LLD problems coded (LRU, Rate Limiter, Parking Lot, Splitwise, BookMyShow, +1)
- ☐ 1 backend MC in 90 min **and** 1 frontend autocomplete
- ☐ Tier 3 LLD Q&A (15 questions) — explain 10 without notes

PHASE 7 – System Design (Days 141–168)

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Goal: Product HLD in 45 min (user + metric + API + bottleneck) → distributed tradeoffs → CS + frontend MC

1 CR bar: HLD sounds like a **product** (Instagram/Uber/Drive), not an infra catalog. Kafka/K8s only if the bottleneck needs them.

7A 🟢 EASY – SD building blocks (Days 141–143)

Difficulty: Concepts + estimation only (no full design yet)

Theory: [10c-system-design-prerequisites-complete.pdf](#) (complete)

Day	Study	Practice
141	Vertical vs horizontal scaling, load balancer algorithms	Draw 3-tier architecture from memory
142	Caching layers (browser → CDN → Redis → DB)	Explain cache-aside vs write-through
143	Read replicas, message queues intro, estimation drills	5 QPS/storage calculations

What success looks like: Back-of-envelope: 10M DAU → QPS in 2 min.

Gate: Draw caching layer diagram + explain 3 LB algorithms before any HLD.

7B 🟢 EASY – Simple HLD (Days 144–147)

Difficulty: Single-service designs (30 LPA–SDE2 level)

Theory: [10-system-design-complete.pdf](#) \$URL Shortener, \$Paste Bin

Interview Q&A: Tier 4 HLD Q1–5

Day	Design	Key concepts	Time
144	URL Shortener	Base62, hash, redirect, Redis cache	45 min draw + explain
145	Paste Bin	TTL, object storage, cleanup job	45 min
146	Rate Limiter (distributed)	Token bucket, Redis, per-IP/user	45 min
147	Redo URL Shortener blind (no notes)	–	45 min

What success looks like: API + DB schema + one diagram in 45 min.

7C 🟡 MEDIUM – Standard HLD (Days 148-152)

Difficulty: Standard SDE2–Senior **product** rounds

Theory: [10-system-design-complete.pdf](#) §Feed, §Notification • open with Product RADIO ([20-1cr](#) §5.1)

Interview Q&A: Tier 4 HLD Q6–12

Day	Design	Key concepts	Time
148	Notification System	Queue, workers, push/email/SMS channels	45 min
149	Twitter Feed (basic)	Fan-out on write vs read, timeline cache	45 min
150	Instagram Feed	Hybrid fan-out, celebrity problem	45 min
151	Reddit / Upvote	Counter, sharding by post_id	45 min
152	Redo Notification System blind	–	45 min

What success looks like: Explain fan-out vs fan-in tradeoff with example numbers.

7D 🟡 MEDIUM-HARD – Complex HLD (Days 153-156)

Difficulty: Senior SDE / 1 CR product HLD bar

Theory: [10-system-design-complete.pdf](#) \$Chat, \$E-commerce, \$Payment, \$Uber

Day	Design	Key concepts	Time
153	WhatsApp Chat	WebSocket, message queue, delivery receipts	60 min
154	E-commerce Order System	Inventory lock, order state machine, ACID	60 min
155	Payment Gateway	Idempotency keys, webhook, saga, reconciliation	60 min
156	Uber (basic)	Matching, geohash, surge – high level only	60 min

What success looks like: Identify consistency vs availability tradeoff per design.

Gate: 6 HLD done before distributed systems theory.

7E 🔴 HARD – Distributed systems (Days 157-160)

Difficulty: Google L5 / Meta E4 distributed round

Theory: [11-distributed-systems-complete.pdf](#)

Interview Q&A: Tier 4 Distributed Q1–8

Day	Study + design extension	Focus
157	CAP + PACELC + consistency models	Quiz: CP vs AP for bank vs social feed
158	Replication + sharding + consistent hashing	Extend URL shortener to multi-region
159	Consensus (Raft), leader election	Why ZooKeeper/etcd for coordination
160	Saga, 2PC, outbox, idempotency	Extend payment design with saga

What success looks like: Explain CAP during partition with real product example.

7F 🟡 HARD – Product HLD deep dives (Days 161-164)

Difficulty: Google / Meta / Uber product design (not Kafka class)

Theory: [10-system-design-complete.pdf](#) §14-18 • [20-1cr-product-company-complete.pdf](#) §5

Reference only: [12-microservices-kafka-complete.pdf](#) — use a queue **inside** a product design if async is required

Day	Design (45-60 min blind)	Product metric	Deep dive
161	YouTube / Netflix	start-play p99 < 2s	CDN, transcode pipeline, ABR
162	Google Drive / Dropbox	zero silent data loss	chunk + checksum, conflict, notify
163	Typeahead	p99 < 50ms	prefix index, hot-prefix cache, typos
164	Ticketmaster seats	zero oversell	hold lock, payment race, idempotency

What success looks like: API JSON + one metric + user-visible fallback. Kafka is a sentence, not the title.

7G 🟡 HARD – CS fundamentals + FE MC + company loops (Days 165-168)

Difficulty: Microsoft / Amazon CS sneak questions + Atlassian FE + loop shape

Theory: [20-1cr](#) §3, §6, §7 • [16-behavioral-senior.pdf](#)

Reference only: [13-kubernetes-cloud-advanced-complete.pdf](#) (deploy story if asked)

Day	Study + practice	Focus
165	CS pack OS + DBMS + Networks (20-1cr \$7)	10 rapid-fire aloud
166	Frontend MC #2 (infinite scroll or tic-tac-toe)	90 min timed
167	Company loop map – pick 2 targets; write round order + 4 STAR tags	Google vs Amazon vs Meta vs Uber (20-1cr \$3, \$8)
168	Phase 7 gate exam	1 product HLD blind (60 min) + 5 CS Qs (15 min) + 2 STAR stories

Phase 7 final gate (must pass ALL):

- ☐ 10 HLD designs on paper including Feed, Chat, Uber, Payment, YouTube, Drive, Typeahead
- ☐ Each HLD opens with user + metric + API (not Kafka)
- ☐ Idempotency + lock applied to Ticketmaster / payment
- ☐ CS pack: 10 rapid-fire without notes
- ☐ 2 frontend widgets timed
- ☐ 8 STAR stories **tagged** to Amazon LP / Google / Meta
- ☐ Tier 4 Q&A: answer 20/50 without notes

PHASE 8 – Interview execution (Days 169–182)

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Goal: Convert knowledge into offers at ~1 CR product companies

1 CR bar: Full loop in the shape of the company you applied to (Google ≠ Amazon ≠ Atlassian)

8A 🟢 EASY – Consolidate + prepare (Days 169–171)

Day	Activity	Time
169	Audit weak areas + OA (S3) : 2 mediums in HackerRank editor (21 §3, 22 §4.2)	2 hrs
170	Resume + LinkedIn + recorded recruiter screen (S2) (22 §4.1)	4 hrs
171	Review Tier 1–3 Q&A + CodeSignal-style set or take-home README (22 S3/S8)	2 hrs

Deliverables: 1-page resume, 2 pinned GitHub projects with architecture diagrams.

8B 🟡 MEDIUM – Partial mocks (Days 172–174)

Day	Mock format	Time
172	1 medium DSA (35 min timed) + 15 min review	1 hr
173	Karat-style (S5) autocomplete API + code or LLD LRU/Rate Limiter (45 min)	1 hr
174	1 easy HLD (URL Shortener, 45 min blind)	1 hr

Use: Pramp, Interviewing.io, or record yourself.

8C 🟡 MEDIUM-HARD – Combined mocks (Days 175-177)

Day	Mock format	Time
175	1 medium DSA + 1 medium HLD (Notification) back-to-back	2 hrs
176	1 medium-hard DSA + 1 LLD (Parking Lot) back-to-back	2.5 hrs
177	Behavioral + bar raiser (\$15): 4 STAR + 15 min "what was your decision?" (22 \$4.7)	1 hr

8D 🔴 HARD – Full loop simulations (Days 178-180)

Day	Full loop (simulate real 80 LPA onsite)	Total time
178	Google/Meta shape: DSA medium (35) → DSA medium-hard + follow-up (45) → product HLD (60)	3 hrs
179	Amazon/Atlassian shape: Backend or frontend MC (90) → LP/values behavioral (45) → LLD or product HLD (45)	3.5 hrs
180	Weakest round redo + hiring-manager 20 min (20-1cr \$8)	2 hrs

Pass criteria: Clear communication, optimal or near-optimal DSA, structured HLD, no silence > 30 sec.

8E HARD – Apply + negotiate (Days 181–182)

Day	Activity
181	Apply to 8 from 22 top-100 list – write each company's family (F1-F12) first – + 10-min team-match pitch (22 §4.9)
182	Tier 5 salary Q&A + 1 CR negotiation range + continue applications. If gates missed → start Phase 9 (20–1cr §11)

Ongoing (Days 169–182): 2 LC/day maintenance, 1 application/day minimum.

Phase 6-8 at a glance (print this)

Phase	Sub	Days	Difficulty	1 CR skill
6A	Easy DSA refresh	113-116	●	Pattern recognition
6B	Easy LLD/OOP	117-119	●	Class design
6C	Medium DSA trees/graphs	120-123	●	BFS/DFS fluency
6D	Medium LLD	124-127	●	45-min LLD pass
6E	Medium-hard DSA	128-132	●	DP + intervals + backtracking
6F	Hard + tagged DSA	133-135	●	Company flavor + follow-ups
6G	Backend + frontend MC	136-140	●	90-min build + autocomplete
7A	SD building blocks	141-143	●	Estimation + caching
7B	Easy HLD	144-147	●	Single-service design
7C	Medium product HLD	148-152	●	Feed, notifications
7D	Medium-hard product HLD	153-156	●	Chat, payments, Uber
7E	Distributed systems	157-160	●	CAP, saga, sharding
7F	Product HLD deep dives	161-164	●	YouTube, Drive, Typeahead, seats
7G	CS + FE MC + loops	165-168	●	Microsoft/Amazon CS + Atlassian FE
8A	Review + resume	169-171	●	Application ready
8B	Partial mocks	172-174	●	Single-round pass
8C	Combined mocks	175-177	●	Two-round stamina
8D	Company-shaped loops	178-180	●	Google vs Amazon onsite

8E	Apply + negotiate	181-182		1 CR offer conversion
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PDFs mapped to sub-phases

Sub-phase	Primary PDFs
6A-6C	08b, 08-dsa-patterns
6B, 6D, 6G	09-llid, 15-machine-coding
6E-6F	08-dsa-patterns, 14-advanced-dsa-hard
7A	10c-system-design-prerequisites
7B-7D	10-system-design
7E	11-distributed-systems
7F	10-system-design §14-18, 20-1cr §5 (Kafka PDF = reference)
7G	20-1cr §3 §6 §7, 16-behavioral (K8s PDF = reference)
8A-8E	17-interview-qa, 20-1cr scorecard, exercise PDF month-06

Exercise PDFs: [month-05-dsa-llid-exercises-complete.pdf](#) follows Phase 6 sub-phases.

Exercise PDFs: [month-06-interview-exercises-complete.pdf](#) follows Phase 7 sub-phases.